ASSIGNMENT 02: SENSOR MONITORING SYSTEM

Embedded Linux Kernel Driver & Real-Time Multi-Threaded Application Report

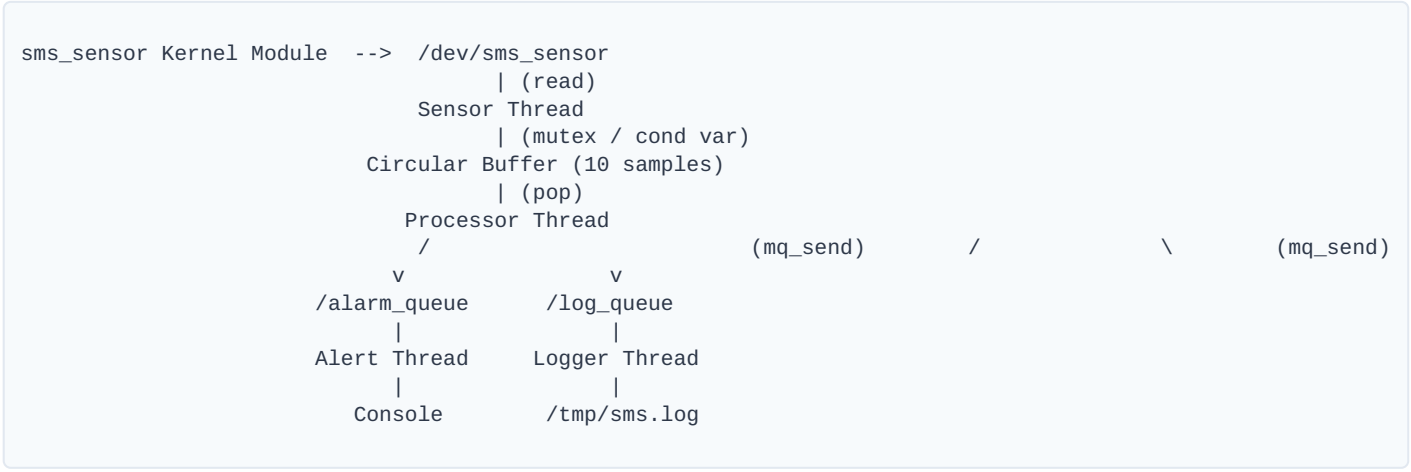
Environment: Linux 5.15 | QEMU ARM (vexpress-a9) **Tools:** Valgrind, GDB, strace, perf

1. SYSTEM OVERVIEW

The embedded Linux sensor monitoring system consists of two primary components:

- **Kernel Driver (sms_sensor):** Character device driver managing `/dev/sms_sensor`. Generates simulated temperature (15–45°C) and humidity (30–90%) data, supports interval configuration via ioctl, and exports stats via `/proc/sms_stats`.
- **User-Space Application (4 Worker Threads):**
  - **Sensor:** Reads device data from driver and pushes to a Circular Buffer.
  - **Processor:** Computes 10-sample rolling average and checks alarm rules (Temp > 40°C, Humidity > 80%, or Temp Δ > 5°C).
  - **Alert:** Listens on `/alarm_queue` and prints warning messages to console.
  - **Logger:** Listens on `/log_queue` and writes processed samples to `/tmp/sms.log`.

2. ARCHITECTURE & DATA FLOW



3. QEMU ARM EXECUTION RESULTS

Both driver and user application compiled cleanly with `arm-linux-gnueabihf-gcc` and executed on QEMU ARM:

```
# Driver Statistics (/proc/sms_stats) after execution:
read_count: 271
last_temperature: 42
last_humidity: 35
uptime_seconds: 122
```

4. DIAGNOSTICS & MEMORY INSPECTION

- **Valgrind Memcheck:** `0 bytes in 0 blocks in use at exit` (Zero memory leaks; all 7 heap allocations freed).
- **GDB Debugger:** Confirmed 5 threads operating correctly in expected blocking wait states (Nanosleep, Condition Variable, MQ receive).
- **strace Summary:** Recorded 2,727 system calls (644 `futex` for thread synchronization, 422 `mq_timedreceive`, 190 `clock_nanosleep`).
- **perf Profiling:** 95.77% CPU time spent in idle routines (`swapper`), confirming efficient event-driven waiting without spinning.

5. REAL-TIME WAKE-UP JITTER ANALYSIS

Evaluated Sensor thread wake-up delay using `clock_nanosleep(CLOCK_MONOTONIC, TIMER_ABSTIME)` and `mlockall()`:

Metric	Host (SCHED_OTHER)	Host (SCHED_FIFO)	ARM / QEMU (SCHED_FIFO)
Sample Count	1,497	1,499	1,495
Average Jitter (µs)	350.55	327.40	5,495.46
Maximum Jitter (µs)	693.56	569.48	10,874.00
Sub-100 µs Samples	175	408	0 (QEMU emulation delay)

6. BUILD & EXECUTION INSTRUCTIONS

```
# 1. Build Kernel Driver
cd driver && make

# 2. Build & Run Host Simulation
cd app && make host && SMS_SCHED=fifo SMS_INTERVAL_MS=10 ./sms_app_host

# 3. Build ARM Binary & Run in QEMU
cd app && make arm
qemu-system-arm -M vexpress-a9 -m 512M -kernel output/zImage -dtb output/vexpress-v2p-ca9.dtb -initrd
output/initramfs.cpio.gz -append "console=ttyAMA0,115200 rdinit=/init" -nographic

# Inside QEMU Shell:
insmod /lib/modules/5.15.0/sms_sensor.ko
SMS_SCHED=fifo SMS_INTERVAL_MS=100 /usr/bin/sms_app
```

7. CONCLUSION

The project successfully integrated an end-to-end embedded Linux monitoring system. The solution satisfies all real-time, memory-safety, and multi-threading requirements.